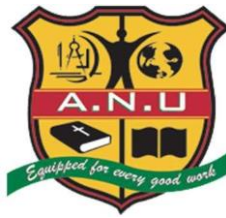


All Nations University



OPTIMIZATION OF BI-DIRECTIONAL OIL AND GAS DRILLING EQUIPMENT USING MATLAB BASED MODULES

A report of the project submitted by

ASIEDU EMMANUEL - ANU18310337

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Bachelor of Engineering

in

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DEPARTMENT OF OIL AND GAS ENGINEERING

ALL NATIONS UNIVERSITY, KOFORIDUA.

OCTOBER, 2022.

DECLARATION

We hereby declare that this submission is our work towards the degree in Bachelor of Engineering in Oil and Gas Engineering, at All Nations University, Koforidua. To the best of our knowledge, it contains no material previously published by another person, nor material that has been accepted for the award of any degree of the university elsewhere, except where due to acknowledgment has been made in the text.

ASIEDU EMMANUEL

ANU18310337

Dr. Charles Boakye

Mr. Desmond Ashikwei

.....

.....

(Head of Department)

(Main Supervisor)

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ABSTRACT

Oil and gas drilling activities have been around for a long time, even with the first drilling activity in 6th century China where drill bits were attached to bamboo poles and transported oil via these bamboo pipelines and used for early lighting and heating applications. Subsurface resources and developments are subjected to drilling wells safely and cost-effectively hence the implementation of recent technology to also increase efficiency and boost production.

In recent times, drilling operations have recorded severe uncertainty in the wellbore position which increases with the depth of the well. This issue affects the overall efficiency of the drill system and reduced the precision of the bidirectional drilling bit.

These uncertainties are caused by downhole vibrations and shocks in the wellbore and these distort processing signals for position monitoring. These signals are measured using the magnetometer and accelerometer sensors mounted in the embedded controls of the drilling system.

Modern research and solutions have been explored but tend to require expensive field tests to the tune of thousands of dollars. This project provides a low-cost approach to utilizing mat lab/Simulink-based models and tools to provide algorithms that help drilling equipment place more precise drill bit and wellbore positions in their reservoirs. This approach takes into account the wellbore environment vibrations and shock signal and hence performs measuring, filtering, and controlling to obtain the desired position of the drill bit and wellbore. Sensor modules are designed in Simulink to filter and convert analog signals from its environment and provide a feedback signal applied to actuators to correct errors and re-position the drill bit.

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LIST OF ABBREVIATIONS

ABBREVIATIONS**STANDS FOR**

VFD	Variable frequency drives
PM	Brushless motor
DC	Direct current
AC	Alternating current
NOV	National Oilwell Varco
PI	Proportional Integral
PID	Proportional Integral Derivative
SPE	Society of Petroleum Engineers
DSATS	Drilling Systems Automation Technical Section
JIP	Joint industry project
IADC	International Association of Drilling Contractors
AUVSI	Association for Unmanned Vehicle Systems International
BOP	Blow out preventer
BHA	Bottomhole assembly is made of stabilizers
MWD	Measurement-while-drilling
LWD	Logging-while-drilling
LQR	Linear Quadratic Controller
MIMO	Multi-input/multi-output

CHAPTER 1

INTRODUCTION

1.0 BACKGROUND

The world's need for energy demand and transformation towards harvesting hydrocarbons is accelerating. This is attributable to a combination of technological progress, development priorities, and environmental concerns. The drilling operation is one of the most sensitive divisions in the oil and gas industry. With geothermal energy as a stable energy resource, deep drilling at attractive costs is pushed every day to the limits. Advancements in the processes, techniques, technologies and innovations associated with well drilling activities all have significant impacts on drilling operational efficiency, safety, and economy, which must be maintained within acceptable levels in unlocking hydrocarbon resources.

Subsurface resource development is subject to drilling wells in safe and cost-efficient manners. Since the discovery of the first oil well in Titusville by Edwin Drake and the progression of the petroleum business over the years, the cable tool and rotary drilling were the first techniques applied in the drilling phase [1]. In recent years, drilling activities for searching hydrocarbons have shown considerable technological advances to obtain safe, environmentally friendly, and cost-effective well construction with increased performance. Some of the early advancements in drilling technology seeking innovative approaches towards better drilling efficiency were based on three basic factors: safety, minimum, and usable hole.

Today's challenge in the field of drilling is to reach deeper targets as fast as possible, minimizing the cost with the continuous improvement of operational efficiency without overlooking health, safety, and environmental guidelines [2]. These factors have rendered drilling activities open to robust techniques and state-of-the-art technologies for an accurate understanding of potential hydrocarbons to be drilled. Improvements in techniques, materials, equipment, autonomous processes, and advanced technologies have played a role in advancing economically recoverable subsurface resources and reducing the environmental impact at the surface and in the underground. Modern drilling technology is categorized as an advanced suite of both surface tools and downhole tools with the capability of drilling entire well sections in one run with far less impact on the surface and subsurface environments [3].

This project explore the use of Matlab Simulink design models for real-time analysis of drill string dynamics. Simulink control strategies are employed to optimize surface drilling parameters concerning bi-directional drilling and reduce vibration-related problems. Although there are complications in accessing the high-frequency downhole data, Matlab comes in handy to obtain close to accurate data which can be used as sample field data.

PROBLEM STATEMENT

1.1 AIM

- To develop a Matlab Simulink model that controls drilling string equipment to optimize system parameters and accuracy of the drill bit.

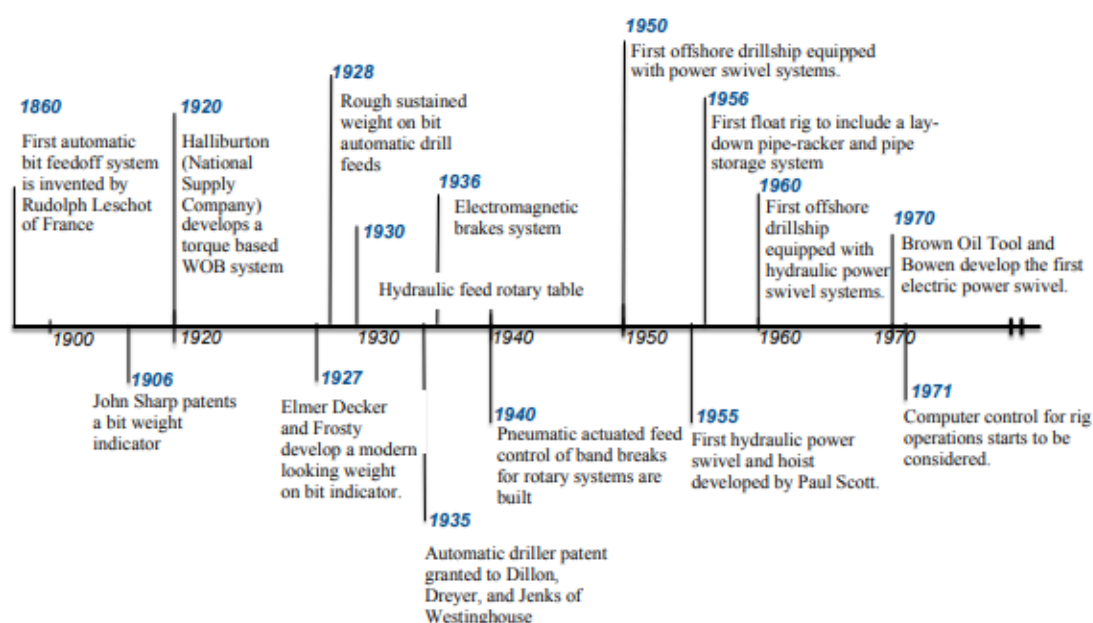
1.2 OBJECTIVES

- To use Matlab Simulink to design and implement control system strategy
- To develop a drill string equipment model using Simulink
- To develop a downhole vibration data

CHAPTER 2

LITERATURE REVIEW

Automation in the oil and gas industry has been aimed mostly toward the mechanization of discrete drilling processes such as rotary speeds, weight on bit, and pipe-handling control. Eustes highlights relevant early work at automating drilling operations. Drilling automation started with the invention of the first rotating diamond bit feed system used for drilling tunnels by Rodolphe Leschot of France in 1860 [6]. The notion of weight on a bit as we know it today was proposed in 1923 as noted by the patent filed by Hild Frederic W in 1927 [7]. However, its importance became popular within the industry until late 1927 as many other inventors and companies started to improve the performance of several draw-works designs which began from rudimentary metallic breaks, replaced by hydraulic-controlled machines followed by pneumatic, and electrical systems. Today's draw-works systems are usually electronically controlled AC or DC motors with a gearbox design, sometimes coupled with an electro-hydraulic system. Recent advances in Variable Frequency Drives (VFD) and Permanent Magnet (PM) or Brushless motor drives [8] has led to the rise of such technologies within the industry due to their improved efficiency, precision, and controllability. This change is univocally beneficial for Drilling Automation as the further reliability and repeatability of an operation the "easier" is to implement an automated system to perform it.



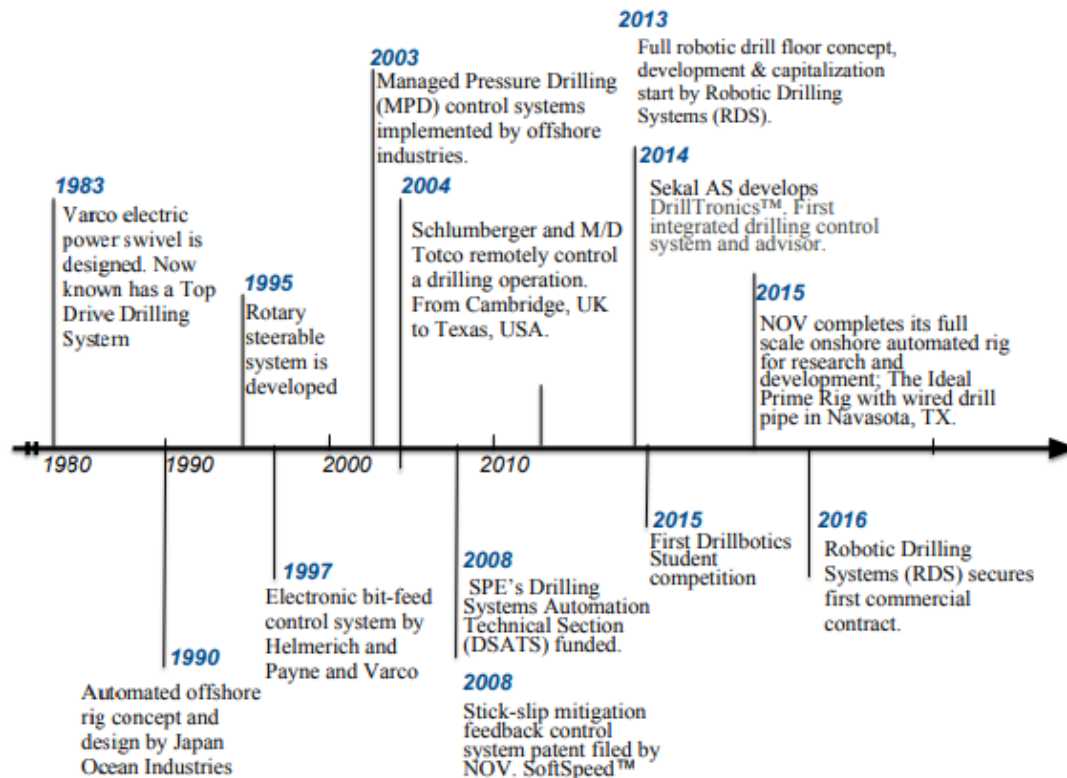


Figure 1 Advances on Drilling Automation [8]

2.1 Top Drive System

The invention of the rotary table brought about one of the most revolutionary technologies in the oil and gas industry, the Top Drive System [9]. The system replaces the Kelly and rotary table of previous rig designs. It is composed of several electric motors coupled with some hydraulic mechanisms to automatically connect stands; modern top drive systems use DC or VFD AC motors, which makes it compatible with advanced controllers to mitigate Stick-slip such as Soft Speed. In addition to providing the rotary force for drilling.

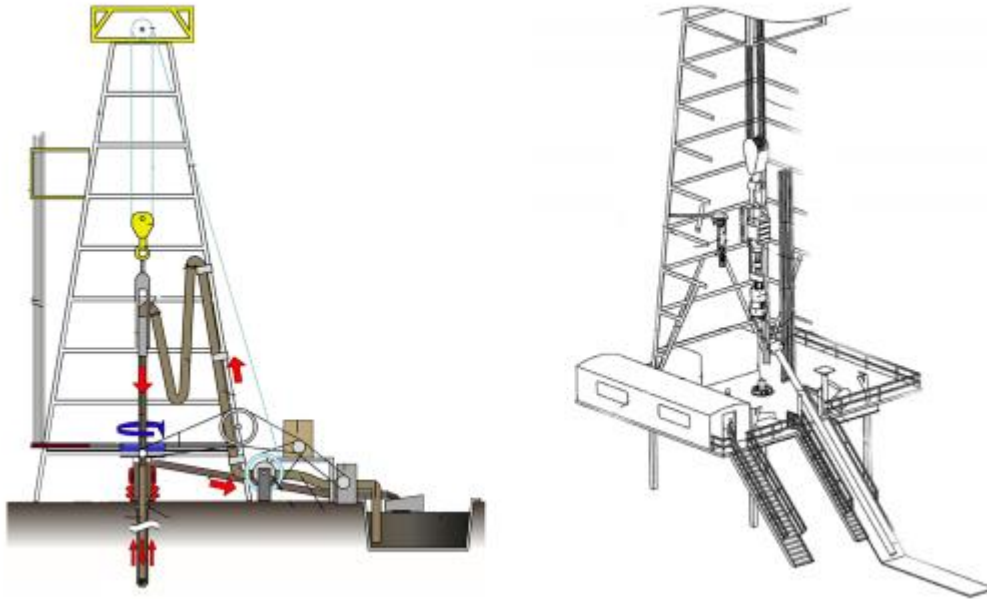


Figure 2 Kelly and rotary table and top drive [9]

Most power electronic devices such as switches and semiconductors over the past decade have considerably become more affordable and dependable; this has driven the introduction of more efficient AC motor controllers such as the variable frequency drive (VFD). Modern Top Drives systems utilize this technology which allows controlling an electric motor behavior precisely by varying the frequency and voltage feed into it. VFD has several advantages; the most important is the availability of accurate real-time feedback information such as Torque, Voltage, and rotation speed without any instrumentation in place [10]. This has led to the development of new control techniques such as NOV Soft Speed to mitigate bit dysfunctions such as stick-slip. [11]. A field case example of the implementation of such a system is shown below. According to NOV's patent by Nessjoen [12]. The system is a closed-loop proportional-integral (PI) controller that 'reacts' to torque oscillations on a limited frequency band typical of the Stick-Slip dysfunction while drilling. A new version of the system released in 2010, is an adaptation of a classic PID (Proportional Integral Derivative) controller. A "PIJ" controller is essentially a classical PI controller with an inertia compensation J-term that is proportional to the difference in measured acceleration only. The J-term increases the controller frequency spectrum to compensate for second mode Stick-slip modes. The system is also capable of auto-tuning the parameters as a function of the drill string length [13].

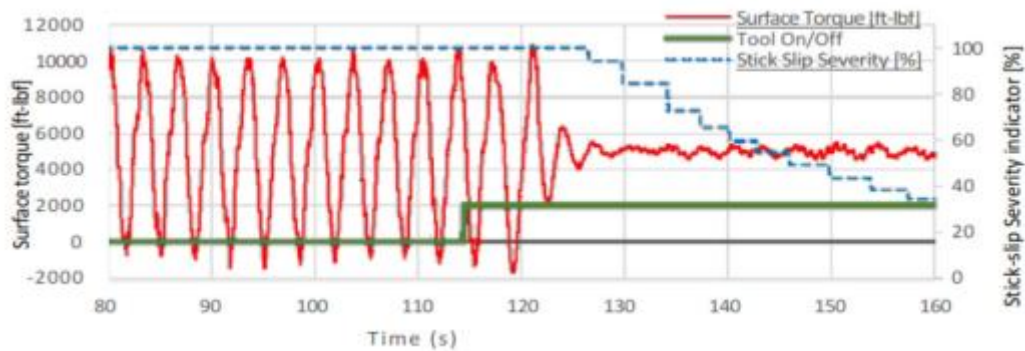


Figure 3 Active torsional dampening stick-slip mitigation system [14].

The Society of Petroleum Engineers (SPE) started an industry technical section focused on Automation technology applied to the drilling industry, the Drilling Systems Automation Technical Section (DSATS) [15]. The purpose of DSATS is to accelerate the development and implementation of automated systems in the Oil & Gas Drilling industry. The group has been one of the most active technical sections within SPE. In the Summer of 2013, DSATS launched the Drilling Systems Automation Roadmap initiative joint industry project (JIP) affiliated with the International Association of Drilling Contractors (IADC), and the Association for Unmanned Vehicle Systems International (AUVSI). The objective is to promote a well-defined system 16 roadmap target for an integrated industrial development to enable the interoperability needed for the implementation of new robust autonomous systems. Some advances in the proposed metadata structure and OPAC standards have been made with the participation of industry operators and services companies such as Shell, NOV, and Baker Hughes tangible, the detailed proposal and updated status are available at de Wardt 2015 [16]. DSATS has been actively encouraging initiatives to promote the participation of automation in the industry. The section has held workshops, forums, lectures, and scientific publications to communicate new technology developments, recommended practices, and the standardization of drilling data nomenclature [17]. DSATS has also drawn the attention of other industries by promoting the inclusion of experts from industries where systems automation is more advanced.

2.2 Percussion or Cable Drilling

Percussion drilling is an oft-used manual drilling technique in which a hammering bit is attached to a long cable that is then lowered into a wide-open hole. The technique is also known as cable drilling, wherein the driller uses a tripod to support the tools. By going back and forth with the bit, the action loosens the soil in the borehole, which is then extracted with the help of a bailer [18]. At intervals, the bit is removed while the cuttings are suspended in water, which is then removed by pumping to the surface. A steel casing helps prevent the hole from collapsing down temporarily and also to protect the hole against contamination of groundwater. The temporary setting is to be removed after a permanent screen and casing are installed [19].

This type of drilling is perfect for unconsolidated and consolidated formations including sand, silt, sandstone, and even gravel. Manual percussion can reach depths of about 25 meters.

2.3 ROTARY DRILLING

Rotary drilling is also one of the most common methods of drilling, especially for digging up exploratory and production wells, which boast depths that exceed five miles below the ground. In this method, lightweight drills are used to drill low-depth wells on land. Rotary mobile and floating drills of various sizes are then used for drilling exploration wells. The equipment is mounted on a platform with a 40-meter-high derrick, that consists of a rotary table, a handy engine, a mud mixer, and an efficient injector pump [20]. It also includes a winch and 27-meter-long pipe sections. The rotary table then directs the square Kelly, which is connected to the drilling pipe. The mud swivel on the pipe is subsequently connected to blowout preventers. The pipe is known to rotate at a velocity of 40 to 250 rpm with the drill made of drag bits, sharp cutting edges, or rolling cutters with strong teeth. Cuttings are then removed by fluid circulation that gets inside the pipe. The penetration rate is faster when using air-based drilling fluids compared to water-based ones. A drag bit is useful in this scenario to penetrate unconsolidated sediments, while the roller bit can drill through consolidated rock. The overall rotation speed of the drill can be increased or decreased depending on the hardness of the formation material. The rotating system allows the rotation of the drill string, and it consists of the rotary table, the Kelly, and the swivel [21]. In modern rigs, a top drive groups together the functions of the above three items of equipment.

2.4 Dual-Wall Reverse-Circulation Drilling

Dual-wall reverse-circulation is a type of rotary drilling itself in which two concentric drill pipes are used to create a controlled flow. The drilling fluid is pumped through an outer swivel to reach the bottom of the bit and then ricochets upward into the main pipe. All cuttings are carried upward through an internal pipe and with the help of a surface swivel [22]. The method also allows for geologic sample collection, with samples usually delivered through the cyclone created at the surface. The method is compatible with both rotary drilling and percussion drilling methods. The fluids are circulated within a controlled space and all cuttings can be recovered at any time. The method applies to all types of geologic formations and does not require surface casing, too. Good sample recovery is one of the main strengths of this method. The method also allows for fast penetration in alluvial or fissured rock. Also, it does help in delivering an accurate estimate of aquifer yield from the depths of the formation [23].

2.5 Electro-Drilling

In this method, rotary tables, winches, and the like are driven by electric motors, thus leading to better flexibility in operations along with remote-controlled drilling. These drills are new methods of oil and gas exploration, as they provide more direct power to the drill bit by connecting the motor above the bit, below the hole. The electro-drilling system has been successful in complex geological conditions where there is the use of weighted mud or mud mixtures. Electro-drilling has made its presence felt in various places including Turkmenistan, Azerbaijan, and Ukraine, which are keen to implement electro-drilling widely to boost savings in energy and material usage [24].

Combining the advantages of rotary and hydraulic-motor methods, this method involves a big range of drill-bit rotational speeds. This method can also use different borehole cleaning agents. One can even implement controlled drilling of deviated boreholes with this method [24].

2.6 Directional Drilling

Directional drilling is an extension of the rotary drilling technique, which guides the drill on a curved path with the deepening of the hole. Directional drilling can help oil explorers reach deposits that cannot be extracted by vertical drilling. Reduced costs are the major driver, as several wells can be drilled in all directions on a single platform [25]. This also allows one to tap into undersea reservoirs and now, by using computers to guide the automatic drilling machines, without connecting and disconnecting sections. Some other drilling methods include abrasive drilling which makes use of abrasive material for driving pressure to cut through the substrate, and explosive drilling and flame piercing, two more methods that are used in some drilling scenarios [25]. At a specified vertical depth, this definition is the fundamental concept of "controlled" directional drilling even in a good bore which is held as close to vertical as possible as well as a deliberately planned deviation from the vertical. Directional drilling is, therefore, a technique that makes it possible to reach deep mining targets even at a considerable horizontal distance from the location of the surface rig. During the drilling process, based on the information as and when it is relayed to the drilling engineer, both the inclination and the direction can still be modified (navigational drilling).

Directional drilling is used in a number of operational situations, the most recurrent applications are as follows;

- Sidetracking,

- Inaccessible Locations,
- Salt Dome Drilling,
- Fault Controlling,
- Multiple Exploration Wells from a Single Well-bore,
- Offshore Multiwell Drilling, Relief Well, Horizontal Wells

Directional drilling can be configured as a Directional or Horizontal well are drilled on the basis of a design that follows precise technical criteria in order to obtain a regular and 'practicable' hole both at the drilling stage and during all of its subsequent productive life. Broadly speaking, the drilling of a directional or horizontal well has the following operating stages. Beginning of the drilling, Deviation of the well, Constant inclination, S-shaped hole (if the design envisages an S-shaped hole to reach the target, or a decrease in the inclination angle to a pre-determined value, at the established depth the inclination of the hole path begins to be decreased by a prefixed gradient – Drop-Off Rate).

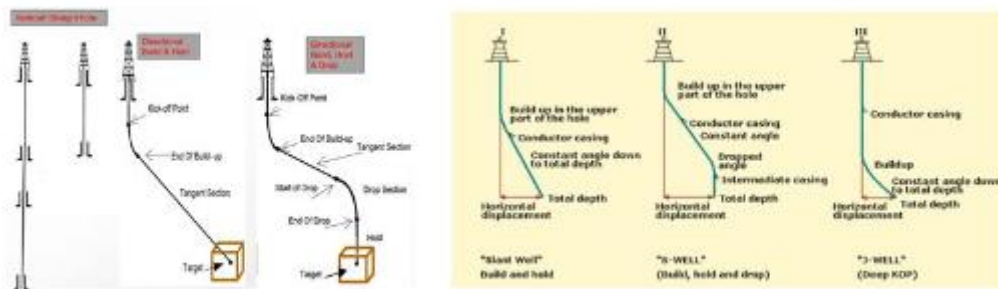


Figure 4 Typical Well Profiles (National Oilwell Varco)

CHAPTER 3

METHODOLOGY

3.1 SIMULATION METHODOLOGY

Oil well drilling is a complex operation and the drilling industry engages the services of personnel and a complicated array of machinery and materials to drill an oil and gas well to depths greater than 6000 meters. The drilling industry has seen technological progress; however, these advances have not changed the fact that, besides the use of complicated machinery, successful drilling is a result of a tremendous team effort. Numerous personnel from the operating company and several service companies work together to drill and complete an oil and gas well. A drilling rig is used to drill a hole, and this requires qualified personnel, different types of equipment the application of a great variety of techniques. For a drilling project to be commenced mainly two goals have to be achieved:

1. The drill and the well must be finished in a safe manner, with no personal injuries or technical problems.
2. The project must be completed with minimum cost.

The overall costs of the well has to be optimized and this optimization may influence where the well is drilled (onshore form an extended reach or offshore just above the reservoir). Either the conventional or slim-hole drilling technology is applied as well as the evaluation procedures run to gather subsurface information for future drilling projects. Rotary drilling is the most efficient technology applied in the oil and gas industry. It is a drilling technology that relies on continuous circular rotation of the bit to break rocks, while drilling fluids circulate through the bit and up the wellbore to the surface, making possible to drill safely and efficiently the well.

3.1.1 Rotary drilling rig

The drilling rig consists of a set of equipment and machinery located on the so-called drilling site and normally the rig is not owned by the oil company but by drilling service companies, which hire out the rig complete with operators and which construct the well according to the client's specifications.



Figure 4 Equipment of a Rotary Drill Rig

1. Crown Block	14. Weight Indicator	27. Degasser
2. Mast	15. Drillers Console	28. Reserve Pit
3. Monkey Board	16. Doghouse	29. Mud Pits
4. Traveling Block	17. Rotary Base	30. Desander
5. Hook	18. Accumulator Unit	31. Desilter
6. Swivel	19. Catwalk	32. Mud Pumps
7. Elevators	20. Pipe Ramp	33. Mud Discharge Lines
8. Kelly	21. Pipe Rack	34. Bulk Mud Components Storage

9. Kelly Bushing	22. Substructure	35. Mud House
10. Master Bushing	23. Mud Return Line	36. Water Tank
11. Mouschole	24. Shale Shaker	37. Fuel Storage
12. Rathole	25. Choke Manifold	38. Engines and Generators
13. Draw works	26. Mud Gas Separator	39. Drilling Line

Table 1 List of equipment

3.1.2 CIRCULATORY SYSTEM

The circulation system consists of mud pumps, distribution lines, and the mud cleaning and accumulation system. It is the closed hydraulic circuit that allows the mud to flow from the surface to the bottom of the hole, inside the drill string, and subsequently back to the surface, in the drill string borehole annulus. The mud from the hole has to have the cuttings removed before being re-injected to the bottom of the hole and the mud pumps supply the energy necessary for circulation; The choice of drilling fluid is dictated mainly by the characteristics of the formations to be drilled, by their drill ability and reactivity to water, and by problems of disposing of the spent fluid. Drilling fluids have many functions to perform including:

1. The removal and transport to the surface of the cuttings produced by the bit;
2. The control of the formation pressure;
3. The prevention of caving and collapse of the borehole walls;
4. The slowing down of the sedimentation of the cuttings when circulation stops;
5. The cooling and lubrication of the drilling equipment;
6. The sources of geological and stratigraphic information.

Drilling fluids are subdivided into three major classes, Water based muds, Oil based muds and Air -based muds (used to reduce bottomhole pressure, to avoid circulation losses in surface layers, or to limit damage to productive formations).

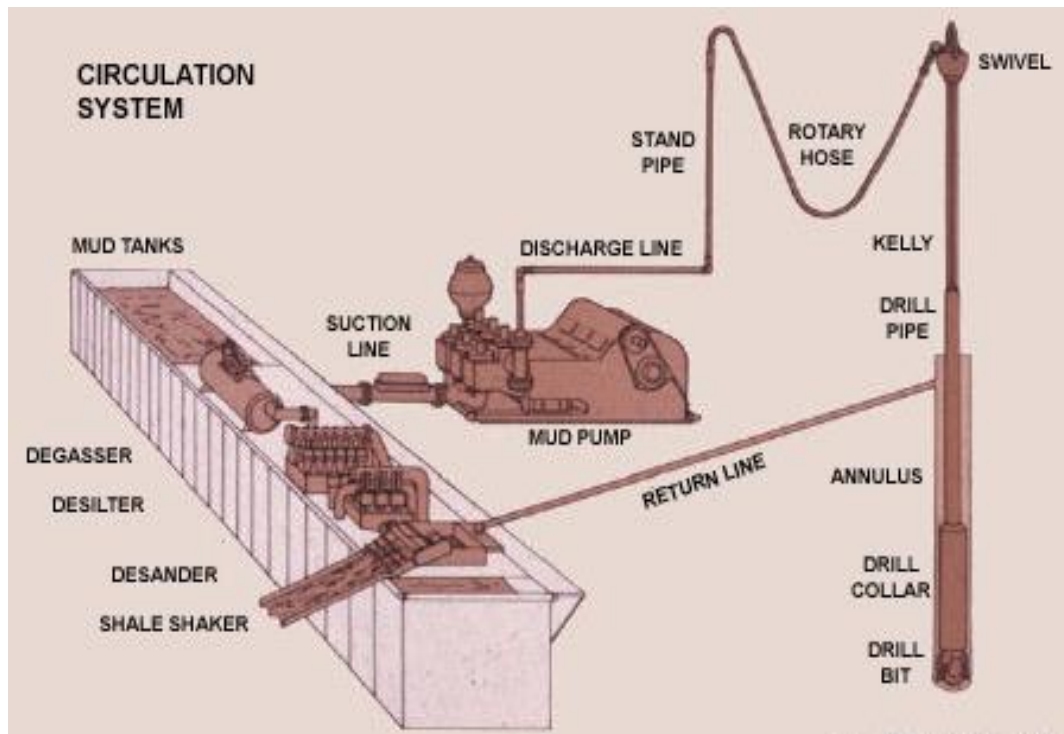


Figure 5 The Circulating System [20]

3.1.5 POWER SYSTEM

A power generation system is needed to run the machines driving the main components of the rig and it is provided by diesel engines, diesel-electric engines. Power is transferred from the engines to the different rig systems by belts, chains, and drive shafts on a mechanical rig, or by generated DC electrical power on an electric rig and it is distributed to the rotary table and mud pumps and to the draw works.

3.1.6 BLOW OUT PREVENTER (BOP)

The well control system or the Blowout Prevention System on a drilling rig is the system that prevents the uncontrolled, catastrophic release of high-pressure fluids (oil, gas, or salt water) from subsurface formations. These uncontrolled releases of formation fluids are referred to as Blowouts. Due to the explosive nature of oil and gas, any spark on the surface can result in the ignition of the fluids and an explosion on the rig. An explosive blowout and the failure of the Well Control System were the causes of the Mocondo Well disaster that killed eleven of the rig crew on the Deep-Water Horizon Rig on April 20, 2010, and resulted in 35,000 to 60,000 bbl/day of crude oil to spill into the Gulf of Mexico.



Figure 9 Blowout Preventer [19]

The blowout preventers are the principal piece of equipment in the well control system and are operated hydraulically; pressurized fluids are used to operate pistons and cylinders to open or close the valves on the BOP. The Accumulators are used to store pressurized, non-explosive gas and pressurized hydraulic fluid to run the hydraulics systems on the rig. The accumulators store enough compressed energy to operate the blowout preventers even if the Power System of the rig is not operating. The blowout preventer is a large system of valves each of which is capable of isolating the subsurface of the well from the rig to provide control over the well. These valves are typically stacked as shown in the figure below and sit below the rig floor on land wells or some offshore well, or they may sit on the seabed on other offshore wells. A schematic diagram of a detailed Blowout Preventer is shown below;

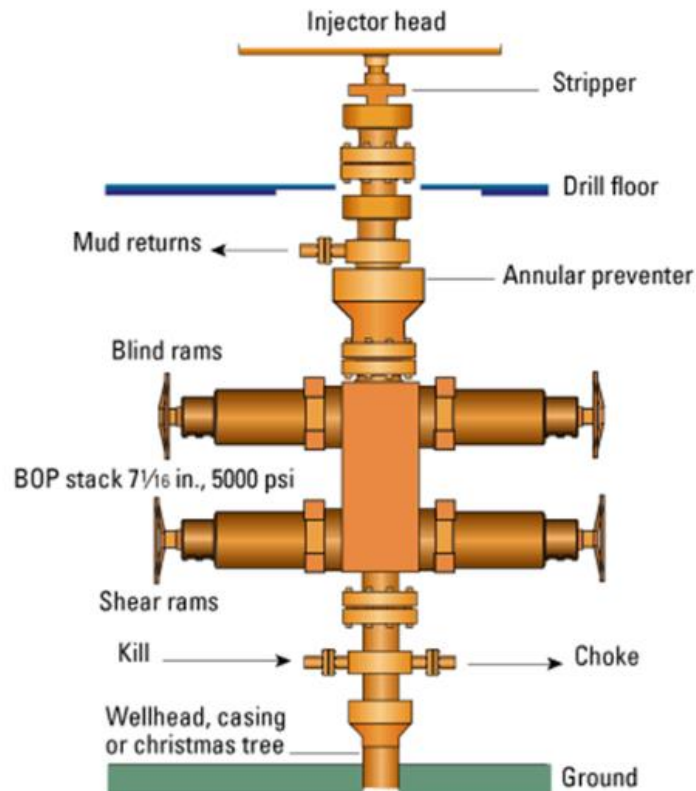


Figure 9 Detailed parts of Blowout Preventer [19]

In the figure above, three types of valves are used, an Annular Preventer, Blind Rams, and Shear Rams. The Annular preventer is the ring-shaped piece of equipment on the top of the BOP. As the name implies, the annular preventer is used to prevent flow through the annular space between the drill string or casing and the annular preventer. The annular preventer can also be used for non-cylindrical pipes, such as the Kelly, or open holes. The annular preventer consists of a doughnut-shaped bladder that when in the open position allows the drill pipe to rotate but in the closed position seals the annulus.

3.2.1 DRILL STRING AND DRILL BIT

The drill string is an assemblage of hollow pipes of circular sections, extending from the surface to the bottom of the hole. It has three functions, it takes the drilling bit to the bottom of the hole, while transmitting its rotation and its vertical load to it, permits the circulation of the drilling fluid to the bottom of the hole, and guides and controls the trajectory of the hole. Starting from the surface, the drill string consists of, a Kelly, drill pipes, intermediate pipes, drill collars, and several accessory items of equipment (stabilizers, reamers, jars, shock absorbers, downhole motors, etc.), and it ends with the bit. The bit is connected to the end of the drill string – it is the tool that bores the rock, transforming it into fragments called cuttings, which are then transported to the surface by the drilling fluid. The choice of the type of bit depends on the hardness, abrasiveness, and drill ability of the rock formation.

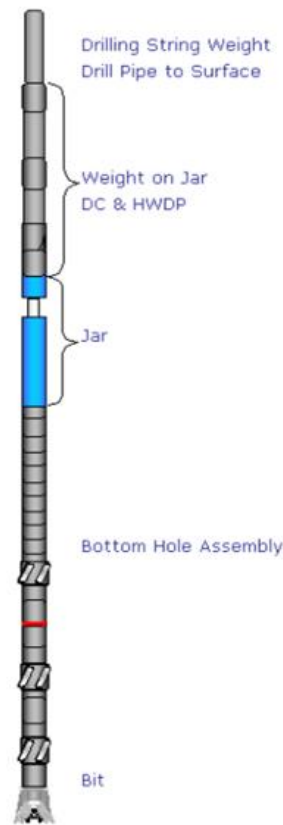


Figure 10 Drill String [19]



Figure 11 Insert Tricone and PDC Drill Bits

3.2.4 DRILLING TOOLS FOR DEVIATION

There are several systems to establish and carry out the deviation and some of them have been highly improved in recent times, particularly since the carrying out of

extended-reach horizontal wells or those having a particular shape have become more common. The drilling industry has gone from using the whipstock and jetting to the systematic use of bottomhole motors, steerable systems and the geosteering. In general, wellbore trajectory is controlled by the type of bottom hole assembly used and the weight on the bit. The bottom hole assembly consists of several components, Heavy-weight drill pipe, Drill collars, Stabilizers and Subs. A typical surface rotated bottomhole assembly (BHA) is made of stabilizers, drill collars and measurement-while-drilling (MWD) equipment. The placement and size of stabilizers control inclination (deviation angle from the vertical). Assemblies can be designed to build angle, hold it steady or drop angle.

3.2.4.1 STEERABLE DRILLING SYSTEM AND GEOSTEERING

Rotary assemblies do not permit close control of wellbore azimuth (compass bearing of the wellbore concerning magnetic North). This control is usually achieved with a downhole motor with a bent housing, which allows the rotation to be applied to the bit alone. Downhole motors are hydraulic machines at the end of the string, screwed directly onto the bit, and the entire mud flow goes through them, part of the mud pressure being converted into rotary motion and torque.

In this way, the rotation necessary for operating the bit is supplied by the downhole motor, while the whole drill string can remain stationary, or may be rotated, if necessary, with the rotary table or the top drive. The use of such motors is essential both for directional drilling and for the application of modern techniques for controlling the vertical trajectory of wells. Downhole motors, an integral part of the BHA, are axial-flow machines of tubular shape and are similar in size to a drill collar. These motors do not form part of the standard equipment of the rig but are hired from service companies, which also supply the personnel specialized in using them and which look after their maintenance. Positive Displacement Motors are rotating, closed-type volumetric machines and their internal architecture is they are Moineau pumps made to operate in the opposite direction, whereby the motor shaft is caused to rotate by forcing the mud through it under pressure. The mud, which is forced past the stator and the rotor, fills these cavities and causes the rotor to rotate continuously, thus causing the rotation of the bit alone.

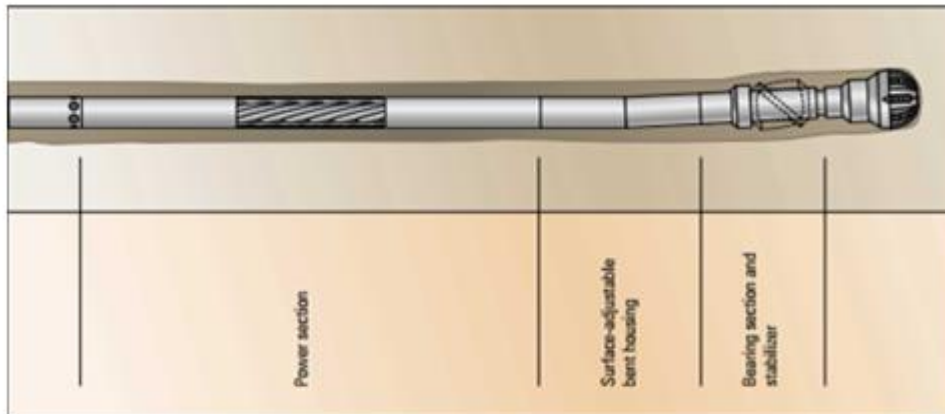


Figure12 Steerable BHA (From Schlumberger)

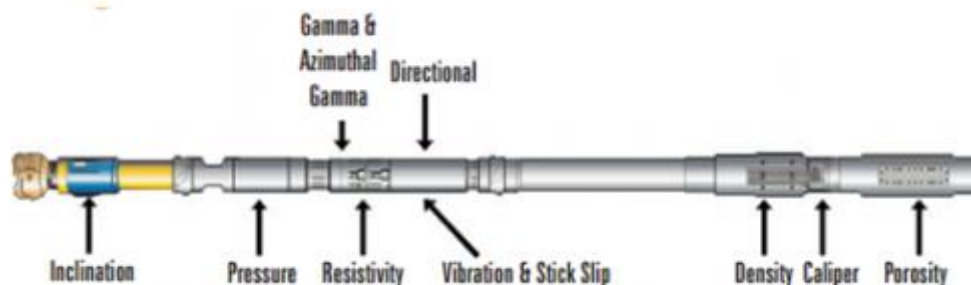


Figure 13 Typical Geosteering BHA (From Baker Hughes)

A steerable drilling system is made up of a PDM, with MWD equipment, which provides in real time the data of interest to the driller (such as inclination, direction, pressure, temperature, real weight on the drill bit, torque stress, etc.) or the LWD equipment is installed. The latter makes it possible to send to the surface, not only the information mentioned above but also geological data (the gamma-ray log, the resistivity, density, sonic logs, etc.). The coupling of sensors providing information on the course of the good trajectory, in real-time and in a continuous way, with logs characterizing the formations from a geological viewpoint, goes under the name of geosteering. This technique makes it possible to navigate, in the true sense of the word, in the subsurface following the most suitable route to reach the prefixed targets.

3.4 BLOCK DIAGRAM

The figure below describes the block diagram of the system's functional design. It begins with the desired position of the drilled bit and string. In real-time operations, the sensor instruments read the current actual position of the bit and string components and update the LQR controller through the e signal from the summing point. The controller initializes its gain value and sends a set voltage to the high-

torque dc motor which in turn positions the drill string and bit. The vibrations signals in the downhole assembly affect the actual position and angle of the drill string and bit. On sensing the difference from the desired position, the LQR controller once again adjusts its control signals to reposition the string and bit in the bottom hole environment.

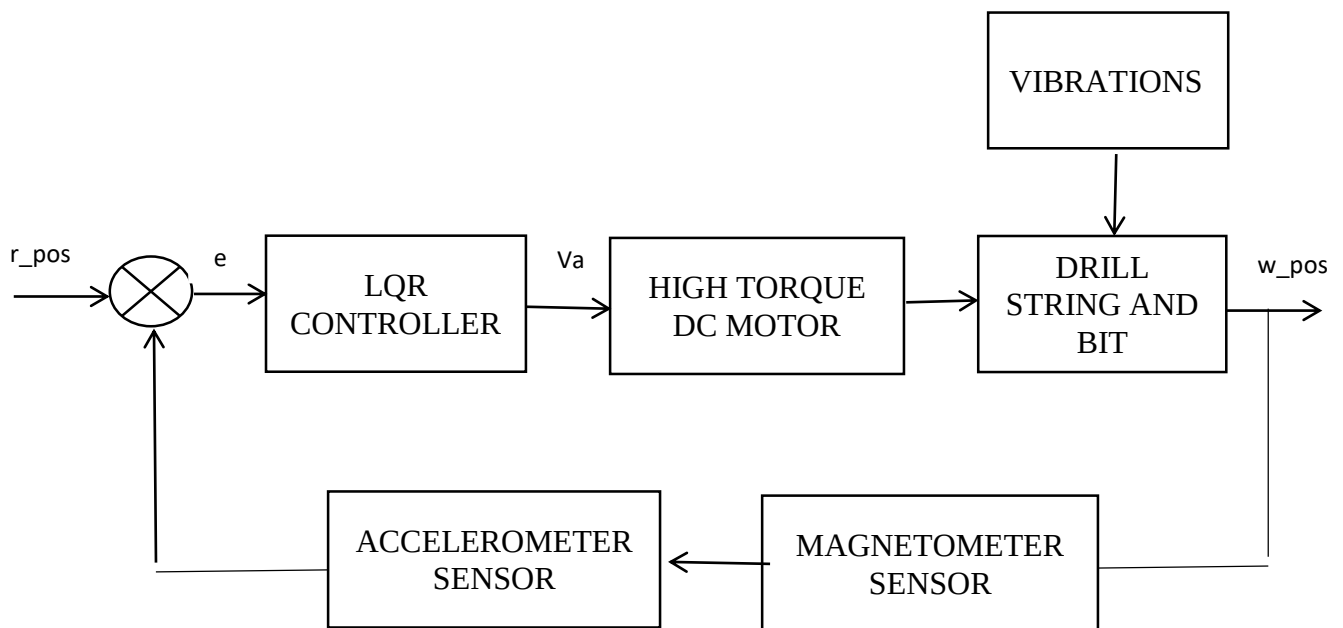


Figure 14 Block Diagram

3.5 CONTROLLER DESIGN

Controller design for coupled multi-input/multi-output (MIMO) systems can be carried out easily for linear plants with well-known linear design techniques, e.g., pole placement. The resulting large matrices can be handled by modern high-end CPUs. However, from a practical point of view, there are some severe drawbacks. A correspondence between system behavior and specific matrix elements often does not exist. Putting the plant into operation is a very difficult task. A change of system structure, caused for instance by a web tear, can result in completely unpredictable and indefinable system reactions. The linear quadratic regulator controller was designed using the Matlab root locus approach. Root locus design is a control system design technique in which you edit the compensator gain, poles, and zeros in the root locus diagram.

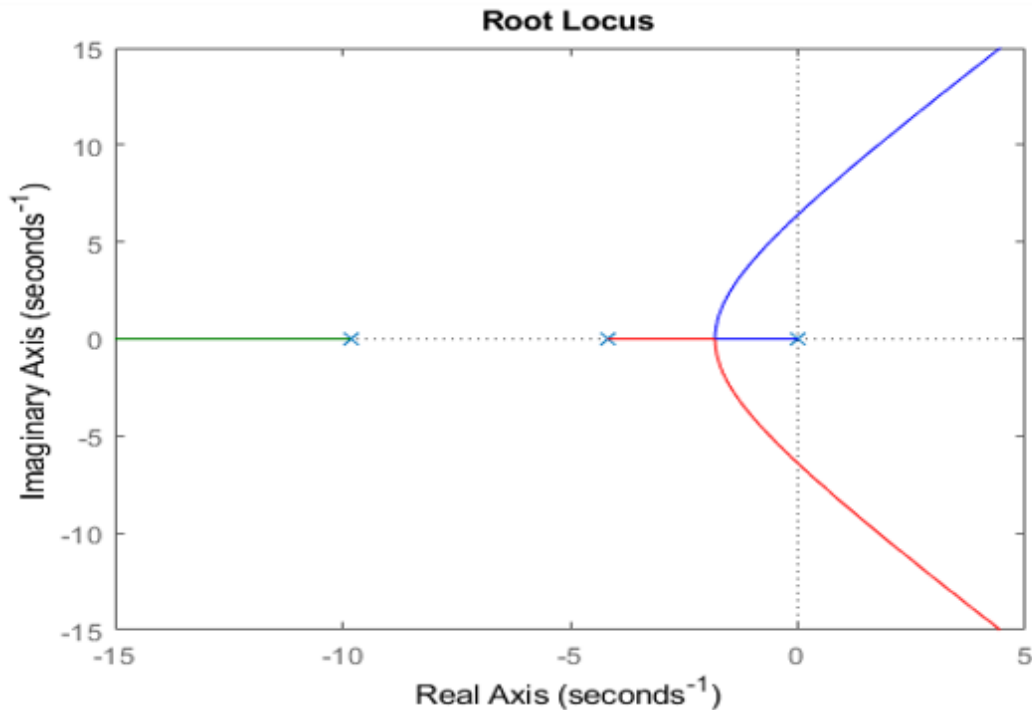


Figure 15 Root Locus Design

The root locus design in Matlab is an interactive procedure that gives a precise estimate of the gain value. A compensator gain of 28 was used in the Matlab algorithm design. The increase or decrease of the gain value affects the position of the poles and zeroes in the figure above. The asterisk in blue indicates the dominant poles in the drill string and bit system and they are used to identify the controllability and stability of the controller system. Mathematically, the Routh-Hurwitz criterion was used to understand the operation of the root locus technique. The number of roots of the characteristic equation with positive real parts (unstable) is equal to the number of changes of sign of the coefficients in the first column of the array. Thus, for the system to be stable, all the coefficients in the first column of the array must have the same sign.

CHAPTER 4

SIMULATION RESULTS AND DISCUSSIONS

4.0 LQR CONTROLLER TEST

Matlab algorithm was developed to implement the LQR controller. The gain was selected interactively and applied to the dc motor block. The LQR technique also uses the state vector $x=(i,w)$ to synthesize the driving voltage of the high torque dc motor. The resulting voltage was in the form

$$V_a = K_1 * w + K_2 * w/s + K_3 * i \dots\dots\dots 1$$

Where i is the armature current

For better disturbance rejection, we use a cost function that penalizes large integral error. The integral expression below describes the cost function

$$C = \int_0^{\infty} (20q(t)^2 + \omega(t)^2 + 0.01V_a(t)^2)dt$$

Where $q(s) = w(s) / s$

The optimal LQR gain for this cost function is computed as follows:

```
dc_aug = [1 ; tf(1,[1 0])] * dcm(1); % add output w/s to DC motor model

K_lqr = lqry(dc_aug,[1 0;0 20],0.01);

P = augstate(dcm);           % inputs:Va,Td outputs:w,x
C = K_lqr * append(tf(1,[1 0]),1,1); % compensator including 1/s
OL = P * append(C,1);        % open loop

CL = feedback(OL,eye(3),1:3,1:3); % close feedback loops
cl_lqr = CL(1,[1 4]);          % extract transfer (w_ref,Td)->w
% This plot compares the closed-loop Bode diagrams for the three DC motor control designs

bodeplot(cl_ff,cl_rloc,cl_lqr);
```

The bode diagram design is an interactive graphical method of modifying a compensator to achieve a specific open-loop response (loop shaping). To interactively shape the open-loop response using Control System Designer, use the bode Editor. In the editor, you can adjust the open-loop bandwidth and design to gain and phase margin specifications. To adjust the loop shape, you can add poles and zeros to your compensator and adjust their values directly in the bode Editor, or you can also use the Compensator Editor.

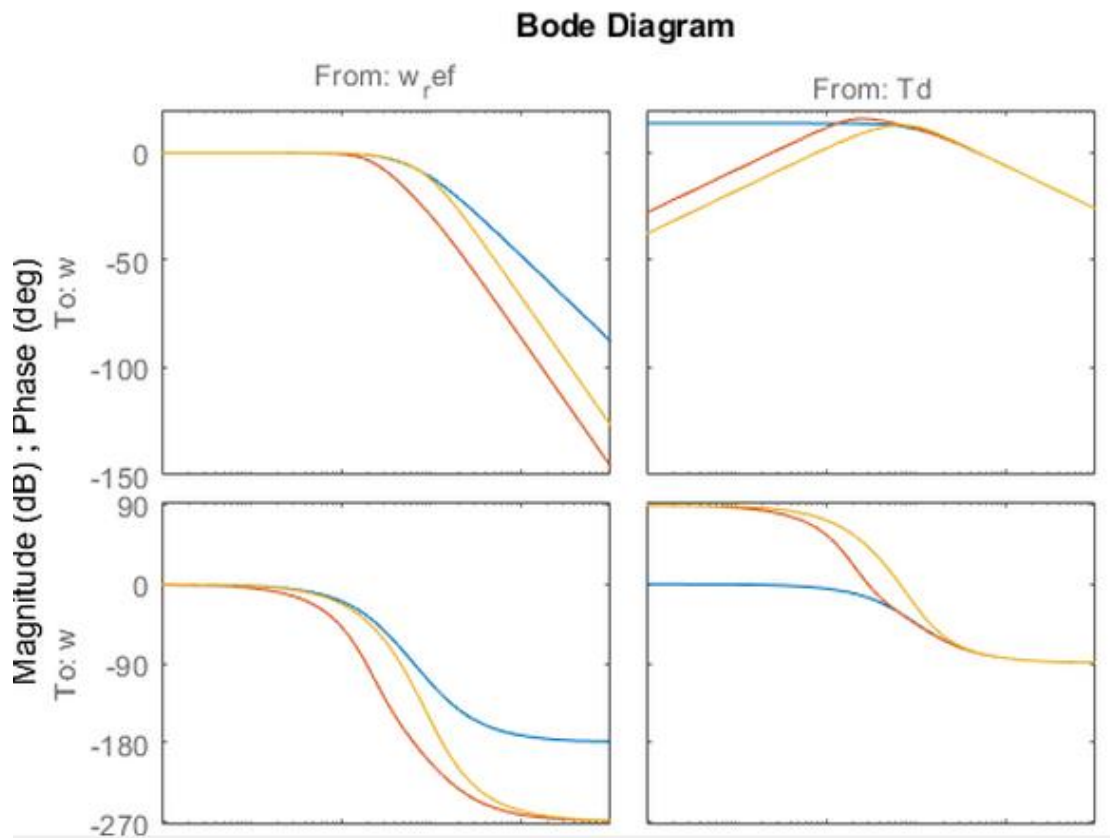


Figure 14 Bode Plot Diagram

4.2 DRILL STRING AND BIT MODEL

The drill string and bit were also developed in Matlab using the mass spring damper approach in Matlab. To observe the behavior of the spring in the bottom hole environment we used the spring's physical properties for the drill string as shown below.

In the figure below the green object is the mass representation of the drill string and bit. It also includes the weight properties of the equipment. The spring connecting the various mass components is based on the spring coefficient which was selected to match the flexibility as well as displacement of the drilled spring. The damping coefficient was also modeled as a second-order system and connected to a fixed ground. Displacement of the drill string is from Z_0 to Z_x where x increases as more string is added to the rig system.

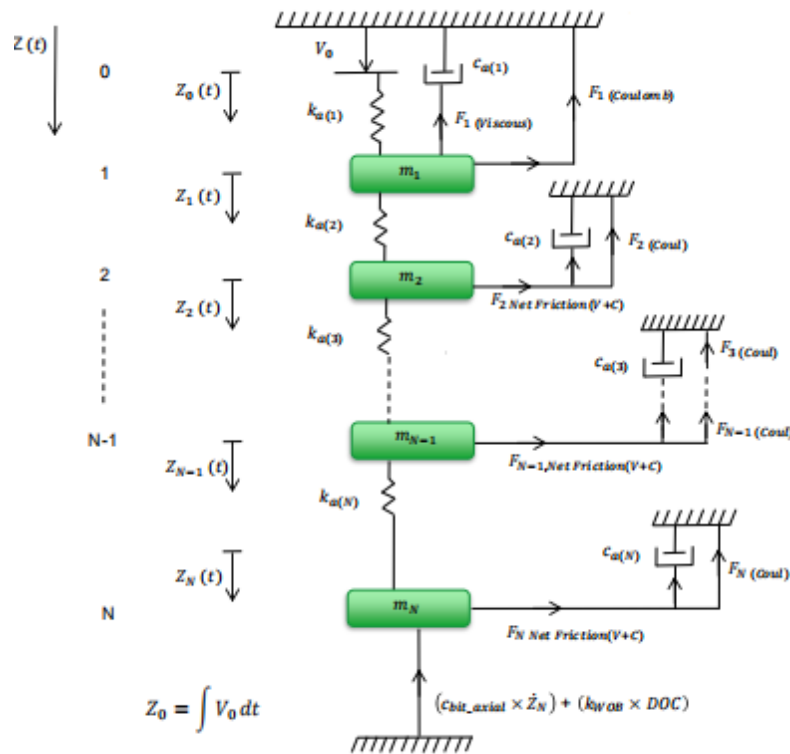


Figure 15 Drill Spring Model

4.2 HIGH TORQUE DC MOTOR TEST

The figure below describes the model developed using Matlab and Simulink. It describes the mud extraction motor interfaced with the drill string and bit. This component extracts the mud collected in the bottom hole area and returns them to the surface top drive. The figure also takes into account the motor angular acceleration and hence the drill bit position can be determined at this point. In figure 17, the drill bit position was simulated. The simulation was carried out over a time interval of 35 seconds and the angular position of the drill bit was recorded. Upon simulation, the drill string and bit oscillated due to environmental factors during drilling and this differs based on the properties at that depth. Oscillations occurred at a range of 15 degrees. Over the course of 35 seconds, the LQR controller dampens the oscillations and sets the drill bit to the desired position at zero.

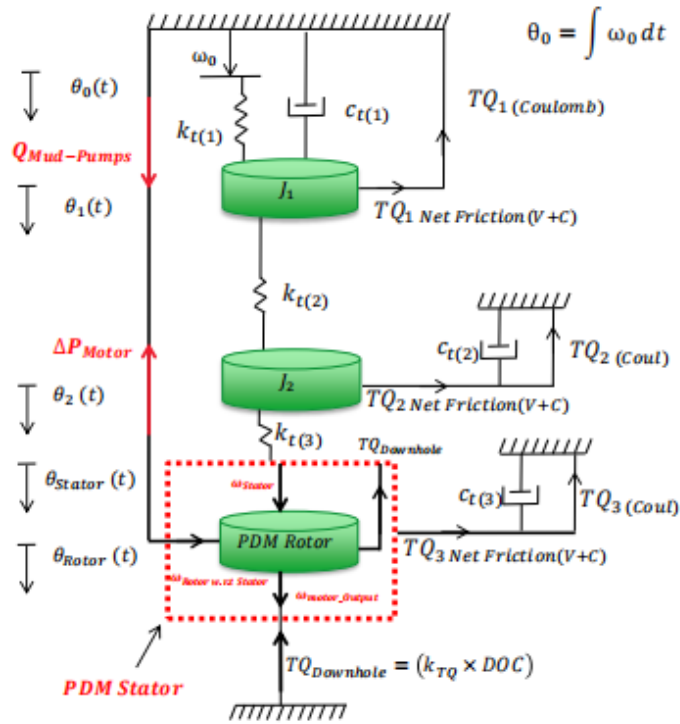


Figure 16 Mud motor model in drill string and bit environment

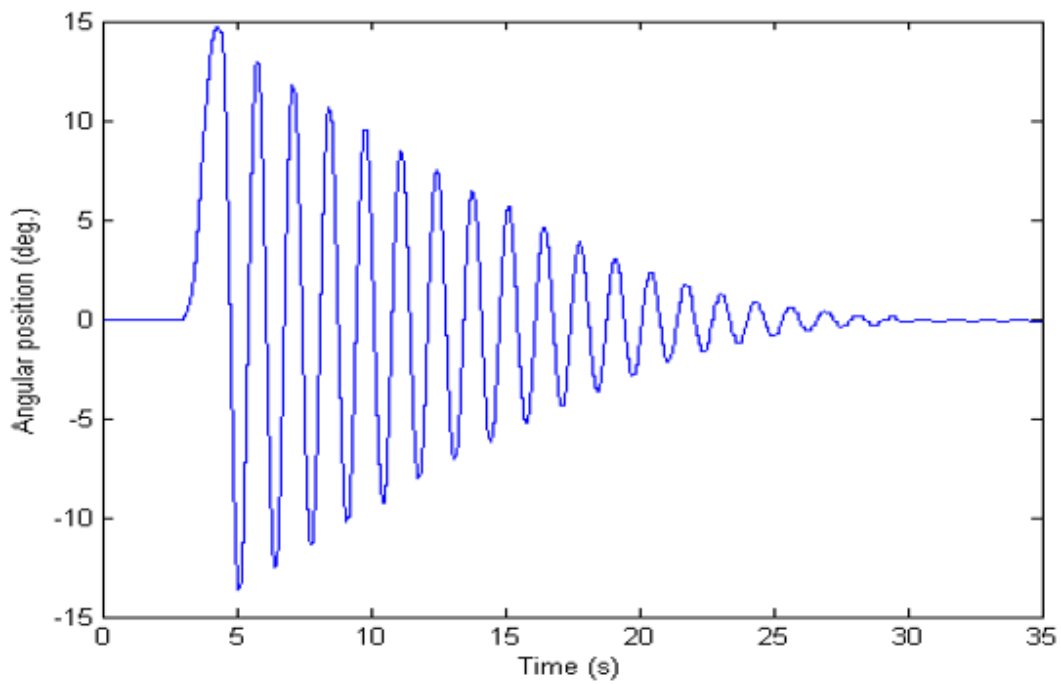


Figure 17 Controlled Drill Bit Position

CHAPTER 5

CONCLUSION

5.0 CONCLUSION

The project was a success with its performance considered acceptable considering the remote nature of field test, cost, and analysis. The LQR controller was designed in Matlab and the desired position was within range of the actual position of the drill string and bit. The simulation was conducted for a period of 35 seconds and from simulations, the drill string and bit oscillated but were quickly damped and controlled within the desired range. The drill string and bit were positioned down to a 0 degree margin which is the desired position of the bit from an initial 15 degree range.

5.1 LIMITATIONS

The design of the system was in no doubt with any limitations. Due to the complex nature of drilling in nature, it comes with factors that limits respective models developed.

- During design of the control algorithm noise was a major factor. The down hole environment had significant vibrations which affected the signal extraction and hence the performance of the overall model.
- For simulation to be concluded a threshold of vibration effects was assumed and this could be different from real field values.
- Due to the remote conditions and depth factors in obtaining real time data, the simulation was largely done using refined data generated from matlab hence may not represent a more accurate environmental situation of the bottom hole assembly.

5.2 FUTURE WORK AND RECOMMENDATIONS

Some recommendations are possible to resolve the limitations as discussed above.

- The algorithm can be improved greatly by utilizing more advance signal processing technologies in Matlab to reduce the noise to signal ratio further.
- More precise dataset should be sourced by conducting more frequent field test on drill bit at the bottom hole environment.

- More advanced controller designs can be explored that provides a more robust system control for the drill string and bit control.

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